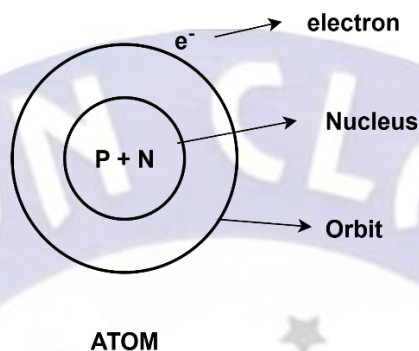


ATOMIC STRUCTURE

- **Introduction**

- Atom is the smallest particle of the matter.
- The word atom is derived from "**Atomos**" (Greek word)

→ Not cut / Indivisible

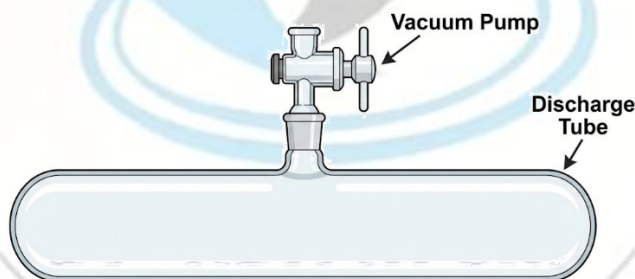


Note:

- Later, Discovered that atom is made by three fundamental particles 'proton', 'neutron' & 'electron'.

- **Discharge tube**

- That tube which has capacity to do low the pressure of gases is called the "discharge tube".

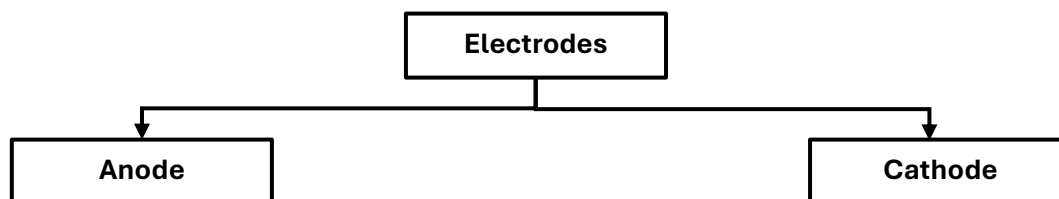


Note:

- Air and mostly gases are electronegative in nature and non – conductor of electricity at room temperature (at 25°C) but at very low temperature it becomes electropositive in nature and conduct electricity.

- **Electrodes**

- The plates / rods which gain by two poles are called electrodes.



1. Anode

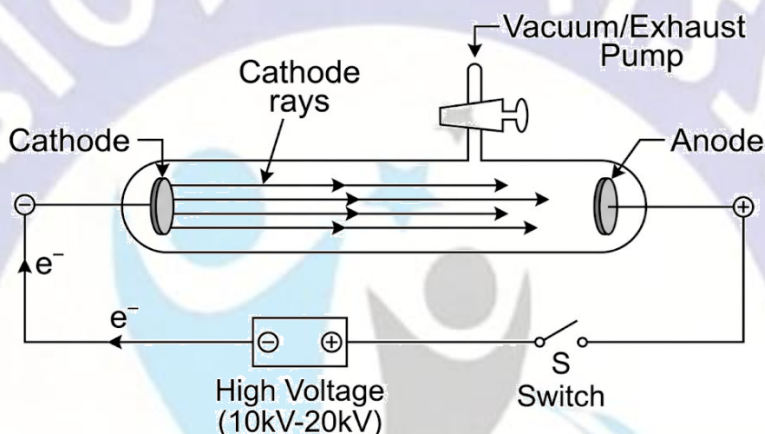
- The positive terminal of any source of electricity is called anode.
- It indicates by positive sign (+).

2. Cathode

- The negative terminal of any source of electricity is called cathode.
- It indicates by negative sign(-).

• Cathode rays

- A beam of electrons in vacuum tube / discharge tube travelling from cathode to anode across the very voltage (10000 V to 20000 V) between the electrodes at very low pressure (0.0001 mm of Hg) at low temperature called cathode rays.
- It is also called electrons beam.
- Cathode rays is discovered by J.J Thomson in 1897.



▪ Properties of cathode rays

1. Cathode rays are emitted from cathode plate to anode plate in the discharge tube.
2. Cathode rays always travel in the straight line.
3. These are a beam of minute material particles.
4. These consist negative charged particles.
5. The cathode rays can effect a photographic plate.
6. The nature of cathode rays does not depend on the nature of the gas in the discharge tube and the nature of the cathode plate material.
7. For each negatively charged particle of a cathode rays. The ratio of charge (e) to mass (m) is constant i.e, e/m ratio is constant.
8. Cathode rays are deflected by electric and magnetic field.
9. When cathode rays are made to strike on a metal it gets heated.
10. Cathode rays can pass through a very thin foil of a metal.
11. Cathode rays forms diffused spots on a photographic plate / film. It is known fogging.
12. When cathode rays strike on the surface of metal like as tungsten copper, etc. then they produce X – ray.

Note:

- The charge (e) to mass (m) ratio (e/m) for the particles of cathode rays obtained from different gases was found to be same which indicates that atoms of all kinds have same type of negatively charged particle. These charged particle is called “electron”.

Thus, electron are the common constituents of all atoms.

i.e, The e/m ratio of particles of cathode rays in $1.76 \times 10^{-11} \text{C/kg}$.

i.e, $(e/m) = 1.76 \times 10^{-11} \text{C/kg}$.

- **Electron**

- Discovered by – J.J Thomson (Joseph John Thomson) in 1897.

- Mass of electron = 9.1×10^{-31} kg
 = 9.1×10^{-28} g
 = 0.000549 a.m.u

- Charge of electron = -1.6×10^{-19} Coulomb

- Relative mass of $e^- = \frac{1}{1836} \times \text{mass of proton.}$

- Mean life of electron $\geq 10^{23}$ years

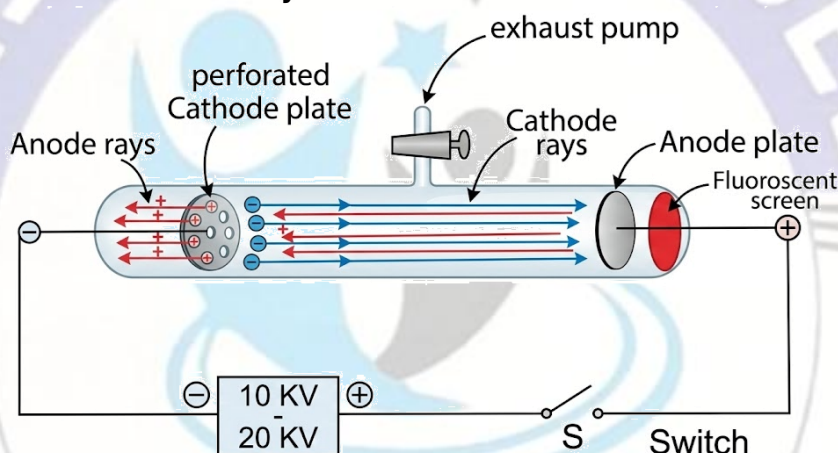
- Electron involves in bond formation.

- Radius of electron $\geq$ radius of nucleus.

→ Slightly greater than

- **Anode rays / Canal rays**

- Anode rays are stream of positively charged particles come out from anode of discharge tube. When a current is passing through a gas at very low pressure then the anode rays passes through canal / whole due to which these are also called canal rays.



- **Properties of anode rays**

1. Anode rays always travels in the straight line.
2. Anode rays consists of positively charged particle known as proton.
3. The nature of anode rays depends upon the gas enclosed in the discharge tube.
4. Anode rays are deflected by electric and magnetic field in the direction opposite to the cathode rays.
5. The ratio of charge (e) to the mass (m) is called e/m value for the particles are not the same for all gases in the discharge tube.

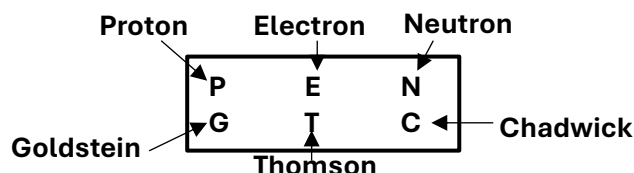
Note:

➤ The e/m ratio for the hydrogen particles is lighter than all gases so, it is called proton.

i.e, Hydrogen – $1e^- \rightarrow H^+$

→ Proton

Trick:



- **Proton**

- Discovered by “Goldstein” in 1919.
- Mass of proton $= 1.6726 \times 10^{-27} \text{ Kg}$
 $= 1.6726 \times 10^{-24} \text{ g}$
 $= 1.00728 \text{ a.m.u}$
- Charge of proton $= 1.6 \times 10^{-19} \text{ Coulomb}$
- Relative mass of proton $= 1836 \times \text{mass of } e^-$
- Mean life of proton $< 10^{23}$ years.
- Proton involves in nuclear reaction.
- It doesn't involve in bond formation.
- Relative mass of proton with respect to neutron is 1.

Special Note:

- The mass of 1 proton is equal to 1 a.m.u

- **Neutron**

- After, discovery of proton and electron. In 1932, James Chadwick discovered that a neutral particle present in the nucleus which is known as “neutron”.

- **Basis of discovery of neutron**

- James Chadwick noticed that all the mass of an atom can not be accounted for on the basis of only protons and electrons.
His observation on the carbon atom (C – 12).

∴ Carbon has 6 protons and 6 electrons.
∴ the mass of electron is negligible.

But the mass of carbon is 12 a.m.u while mass of 6 proton is 6 a.m.u then Chadwick explained this extra mass of 6 units.

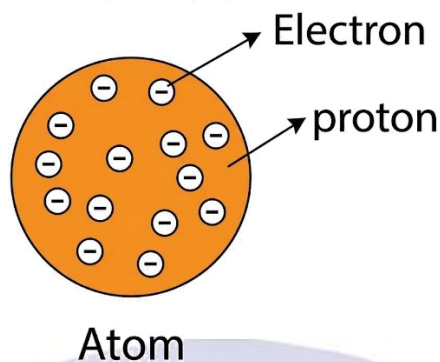
He discovered another subatomic particle present in the nucleus which is equal to proton in mass and neutral in charge. Which is denoted by ‘n’.

For the discovery of neutron he was awarded the Nobel prize in physics in 1935.

- Mass of neutron $= 1.6748 \times 10^{-27} \text{ Kg}$
 $= 1.6748 \times 10^{-24} \text{ g}$
 $= 1.00796 \text{ a.m.u}$
- Charge of neutron = chargless (0).
- Meanlife of neutron = 16.8 years
- It appears in nuclear reaction.
- Mass of proton = mass of neutron
i.e, $M_p = M_n$

- **J.J Thomson atomic model, OR Watermelon theory OR, plum pudding model**

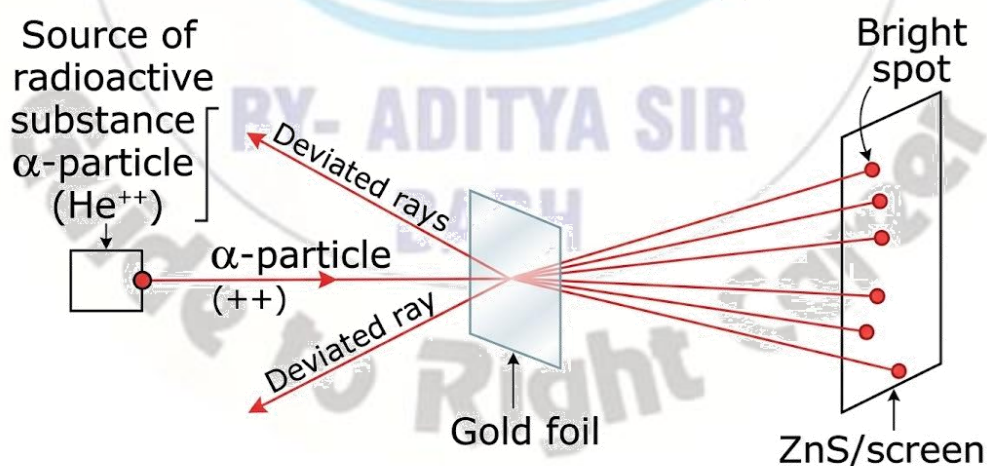
- In 1903, Joseph John proposed his model of an atom on the knowledge of only proton and electron.
- **According to J.J Thomson atomic model**
- An atom has equal magnitude of positive and negative charge, due to which atom is electrically neutral i.e, atom has no overall positive / negative charge.
- The concept of an atom is based on watermelon in which eatable red part is known as proton and embedded black seed is known as electron.



- Thomson assumed that, electrons when disturbed then they start to vibrate from their equilibrium position.
- **Rutherford gold foil experiment**
- In Rutherford's Gold foil experiment, he passes alpha particle from the gold foil.
- In this experiment, Rutherford used a very thin sheet of gold known as gold foil and passes alpha particle through the gold foil.

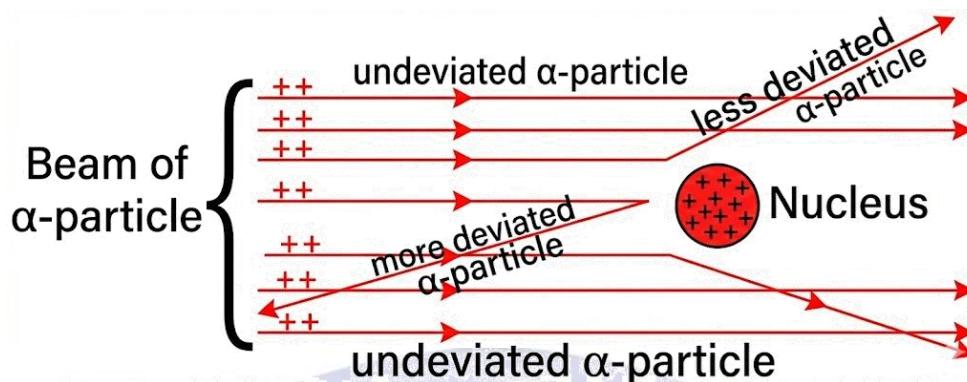
Note:

1. When helium is doubly ionised then we get He^{++} which is known as " α - particle",
→ Charge - 2 units, mass - 4 units
2. The α - particle has amount of energy and it can penetrate through matter to some extent.



Gold foil α -particle experiment

- In above experiment we understand by below picture.



▪ **According to this experiment, Rutherford concluded that**

1. Most of the α – particles pass straight through the gold foil without suffering any deflection from their original path.
2. A few of them (1 in 1 lakh) were deflected through small angles while a very few deflected to large angle ranging from 90° to 180° .
3. Since α – particles which are comparably denser were deflected by the nucleus, it shows that almost the complete mass of atom must be within the nucleus. i.e, mass of the atom centroid in the nucleus.

Note:

- Rutherford discovered the nucleus of atom by this experiment.

▪ **Rutherford explained these observations in the following way**

1. Since most of the α – particles pass straight through the gold foil without any deflection, it shows that there is a lot of empty space in the atom.
2. The observation that some of alpha particles are deflected through small and large angles shows that there is a “centre of positive charge” in the atom which repels the positively charged α – particles and deflects them from their original path.
3. The observation that a very few α – particles completely rebound on hitting the gold foil shows that the nucleus is very dense and hard which does not allow the alpha particles to pass through it.

▪ **An important information about nucleus of atom**

1. Nucleus of an atom is positively charged.
2. Nucleus of atom is very dense and hard.
3. Nucleus of an atom is very small as compared to the size of the atom as a whole.

Note:

1. The nucleus is a small positively charged part of the centre of an atom.
2. Proton and neutrons taken together are known as nucleons.
3. The volume of nucleus of an atom is very small as compared to the volume of the extra nuclear part of the atom.

• **Rutherford's atomic model**

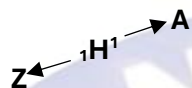
- On the basis of α – particles scattering experiment, Rutherford gave a nuclear model of an atom.

▪ **Rutherford's atomic model describe as follows**

1. An atom consist of two parts these are nucleus (contain proton and neutron) and extra nuclear portion (contain electron).
2. The electrons are distributed in the empty space of the atom around the nucleus in different concentric circular paths, called orbits.

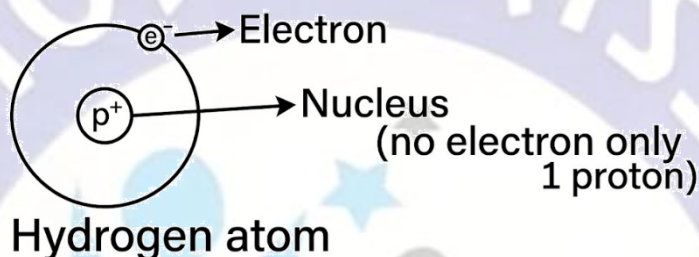
- Note:**

i.e.,

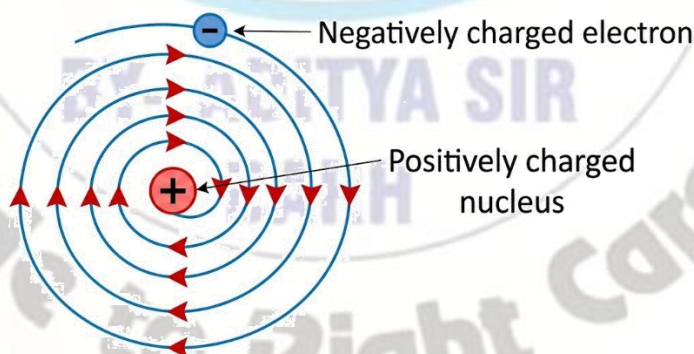


$$\therefore n = A - Z = 1 - 1 = 0$$

OR,



1. The major drawback (defect) of Rutherford's atomic model of the atom is that it does not explain the stability of the atom.
2. According to Rutherford's atomic theory, electrons continuously loses its energy in moving around the nucleus and at the last drop in the nucleus and atom will be destroyed.
continuously loses energy due to which, the distance between orbit and nucleus decreases.

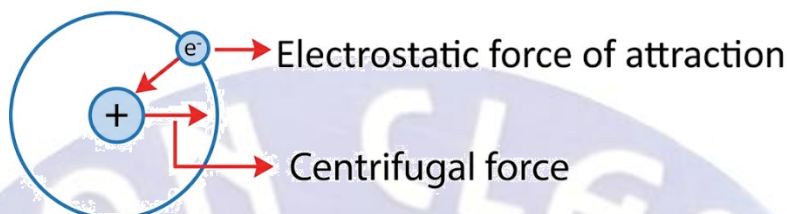


- **Bohr's atomic model**

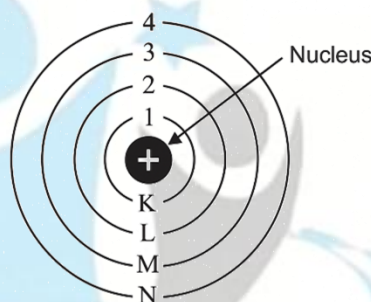
- In 1913, Neils Bohr proposed a modified model of atomic structure, this is known as “The Bohr’s model of the atom”.

▪ **This model based on the following postulate.**

1. In the limit number of orbit, electrons can revolve with fixed amount of energy.
2. An atom is made-up of three fundamental particles like an electron, proton and neutron. These are also called subatomic particles.
3. There is energy limitation of number of electrons.
4. The protons and neutrons are placed at the centre of an atom known as nucleus.
5. Both the electron and nucleus are counterbalanced electrostatic force of attraction and centrifugal force



6. Electrons revolve around the nucleus in the fixed orbit is called stationary orbit.
 7. These circular orbits are also known as energy level or, energy shell.
 8. These orbit are denoted by K, L, M, N, O... OR, 1, 2, 3, 4, 5,...
- The numbering is based on energy level.



Shell no(s)	Shell
1	K
2	L
3	M
4	N
5	O

9. The order of energy level of the these shell: $K < L < M < N < O$.
 $\Rightarrow 1 < 2 < 3 < 4 < 5$

10. When an electron jumps from a lower energy level to a higher one some energy is observed. While some energy is emitted when the electron jumps from a higher energy level to a lower one.
11. Orbits are counted towards the nucleus.

Note:

- Electrons in an atom are not stationary only the energy of electron is stationary in the energy level (shell).

Address of a person	Address of electron
Country name	Shell
State name	Sub – shell
City name	Atomic orbital
House number	House of electron

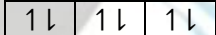
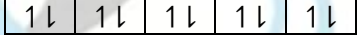
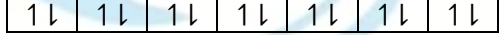
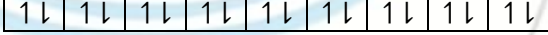
You can find an electron according to above address.

• **Shell / Orbit / Energy level**

Shell	Maximum no. of electrons = $2n^2$
n = 1 (K)	$2 \times 1^2 = 2$
n = 2 (L)	$2 \times 2^2 = 8$
n = 3 (M)	$2 \times 3^2 = 18$
n = 4 (N)	$2 \times 4^2 = 32$
n = 5 (O)	$2 \times 5^2 = 50$

Sub – shell	Full Form	Shape	Maximum no. of e ⁻ (s) = $4n - 2$
s	Sharp		$4 \times 1 - 2 = 2$
p	Principal	 Doubel	$4 \times 2 - 2 = 6$
d	Diffuse	 Double Doubel	$4 \times 3 - 2 = 10$
f	Fundamental	Complex shape	$4 \times 4 - 2 = 14$
g	generate	Complex shape	$4 \times 5 - 2 = 18$

• **Atomic Orbital**

Sub – shell	Atomic orbital
s	
p	
d	
f	
g	

Note:

- According to Hunds, electron first, filled in clock – wise (↑ OR ↓) then anticlock wise (↓ OR ↑).
- If five electrons in p sub – shell then we will fill.

↑↓	↑↓	↑↓
----	----	----

- ↑ / ↓ → Representation of electron.

Special Note:

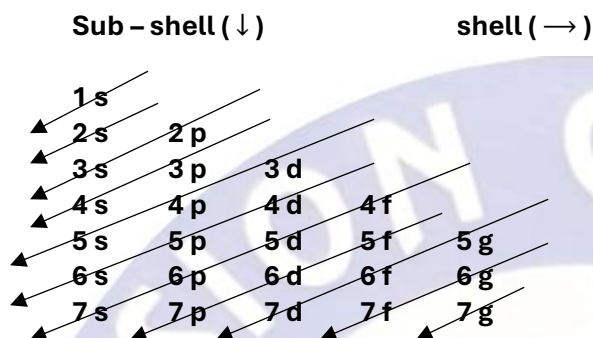
Shell no.	No. of sub – shells	Maximum no. of e ⁻ (s)
1 = K	s	2
2 = L	s + p	2 + 6 = 8
3 = M	s + p + d	2 + 6 + 10 = 18
4 = N	s + p + d + f	2 + 6 + 10 + 14 = 32
5 = O	s + p + d + f + g	2 + 6 + 10 + 14 + 18 = 50

- **Electronic configuration**

- An arrangement of electrons in shell and subshell is known as electronic configuration.
 OR

An arrangement of all electrons in shell and sub – shell of an atom is called electronic configuration of that atom.

- An arrangement of electrons in an atom.



On arranging with full fill electrons

$1s^2 < 2s^2 < 2p^6 < 3s^2 < 3p^6 < 4s^2 < 3d^{10} < 4p^6 < 5s^2 < 4d^{10} < 5p^6 < 6s^2 < 4f^{14} < 5d^{10} < 6p^6 < 7s^2 < 5f^{14} \dots\dots\dots$

Note:

Maximum no. of e⁻(s)
 1 s²
 Shell ← sub – shell

- **Electronic Configuration of atom**

1.	${}_1\text{H}$	$1s^1$
2.	${}_2\text{He}$	$1s^2$ [Full filled]
3.	${}_3\text{Li}$	$1s^2, 2s^1$
4.	${}_4\text{Be}$	$1s^2, 2s^2$
5.	${}_5\text{B}$	$1s^2, 2s^2 2p^1$
6.	${}_6\text{C}$	$1s^2, 2s^2 2p^2$
7.	${}_7\text{N}$	$1s^2, 2s^2 2p^3$
8.	${}_8\text{O}$	$1s^2, 2s^2 2p^4$
9.	${}_9\text{F}$	$1s^2, 2s^2 2p^5$
10.	${}_{10}\text{Ne}$	$1s^2, 2s^2 2p^6$ [Complete Octet]
11.	${}_{11}\text{Na}$	$1s^2, 2s^2 2p^6, 3s^1$
12.	${}_{12}\text{Mg}$	$1s^2, 2s^2 2p^6, 3s^2$
13.	${}_{13}\text{Al}$	$1s^2, 2s^2 2p^6, 3s^2 3p^1$
14.	${}_{14}\text{Si}$	$1s^2, 2s^2 2p^6, 3s^2 3p^2$
15.	${}_{15}\text{P}$	$1s^2, 2s^2 2p^6, 3s^2 3p^3$
16.	${}_{16}\text{S}$	$1s^2, 2s^2 2p^6, 3s^2 3p^4$
17.	${}_{17}\text{Cl}$	$1s^2, 2s^2 2p^6, 3s^2 3p^5$
18.	${}_{18}\text{Ar}$	$1s^2, 2s^2 2p^6, 3s^2 3p^6$ [Complete Octet]
19.	${}_{19}\text{K}$	$1s^2, 2s^2 2p^6, 3s^2 3p^6, 4s^1$
20.	${}_{20}\text{Ca}$	$1s^2, 2s^2 2p^6, 3s^2 3p^6, 4s^2$
21.	${}_{21}\text{Sc}$	$1s^2, 2s^2 2p^6, 3s^2 3p^6, 4s^2, 3d^1$
22.	${}_{22}\text{Ti}$	$1s^2, 2s^2 2p^6, 3s^2 3p^6, 4s^2, 3d^2$

but	$_{22}\text{Ti}$	$1s^2, 2s^2 2p^6, \underline{3s^2 3p^6 3d^2}, 4s^2$ → Shell – 3 – together					
23.	$_{23}\text{V}$	$1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^3, 4s^2$ [Improved]					
24.	$_{24}\text{Cr}$	$1s^2, 2s^2 2p^6, 3s^2 3p^6 4s^2, 3d^4$ [Ground State]					
but	$_{24}\text{Cr}$	$1s^2, 2s^2 2p^6, 3s^2 3p^6, \underline{3d^5}, 4s^1$ [Excited State] → Half filled <table><tr><td>1↓</td><td>1↓</td><td>1↓</td><td>1↓</td><td>1↓</td></tr></table>	1↓	1↓	1↓	1↓	1↓
1↓	1↓	1↓	1↓	1↓			

Note:

- According to Hund's Rule, When d sub – shell is outer sub – shell and has either 4/9 electrons then it accepts 1 electron from its neighbour sub – shell 's' for half filled (Half stability) and full filled (full stability).

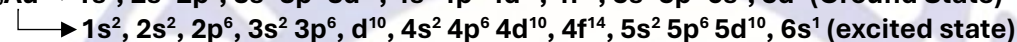
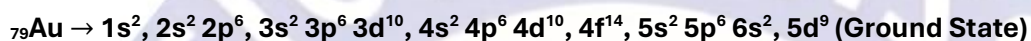
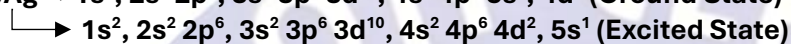
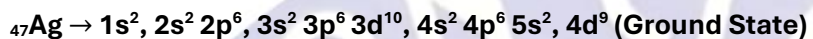
25.	$_{25}\text{Mn}$	$1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^5, 4s^2$
26.	$_{26}\text{Fe}$	$1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^6, 4s^2$
27.	$_{27}\text{Co}$	$1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^7, 4s^2$
28.	$_{28}\text{Ni}$	$1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^8, 4s^2$
29.	$_{29}\text{Cu}$	$1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^9, 4s^2$ [G.S.]
But	$_{29}\text{Cu}$	$1s^2, 2s^2 2p^6, 3s^2 3p^6 \underline{3d^{10}}, 4s^1$ [E.S.] → Full Filled
30.	$_{30}\text{Zn}$	$1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^{10}, 4s^2$

• **Valence electron**

- The outer most shell electrons of an atom is called valence electrons of that atom.

Element	Electronic Configuration					Valence electron
	K	L	M	N	O	
Hydrogen	1					1
He	2					2
Li	2	1				1
Be	2	2				2
B	2	3				3
C	2	4				4
N	2	5				5
O	2	6				6
F	2	7				7
Ne	2	8				8
Na	2	8	1			1
Mg	2	8	2			2
Al	2	8	3			
Si	2	8	4			4
P	2	8	5			5
S	2	8	6			6
Cl	2	8	7			7
Ar	2	8	8			8
K	2	8	8	1		1

Ca	2	8	8	2		2
Sc	2	8	9	2		1, 3
Ti	2	8	10	2		2, 4
V	2	8	11	2		2, 5
Cr	2	8	12	2		2, 6
Mn	2	8	13	2		2, 7
Fe	2	8	14	2		2
Co	2	8	15	2		2
Ni	2	8	16	2		2
Cu	2	8	17	2		2
Zn	2	8	18	2		2



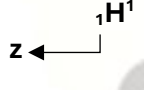
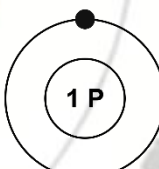
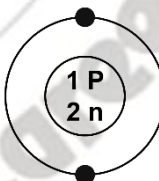
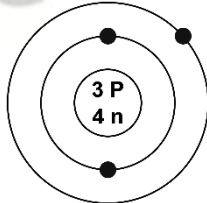
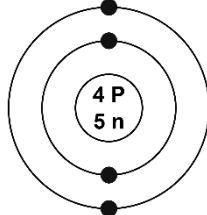
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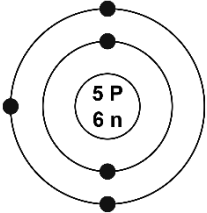
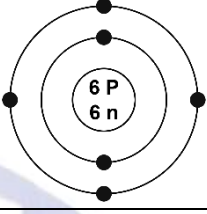
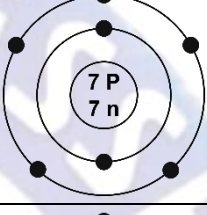
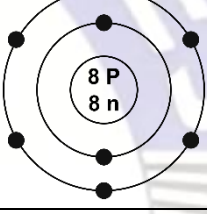
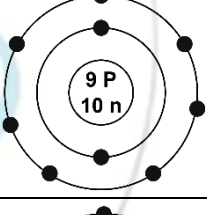
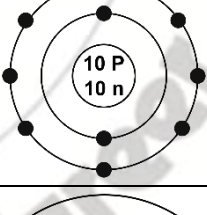
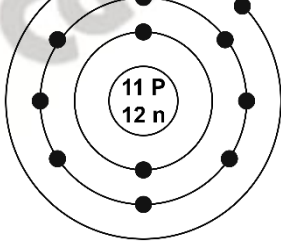
- Valence e⁻(s) take part in chemical reaction.

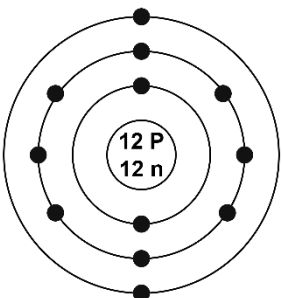
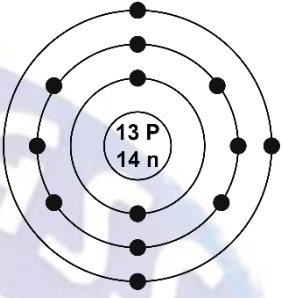
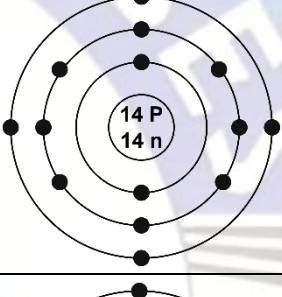
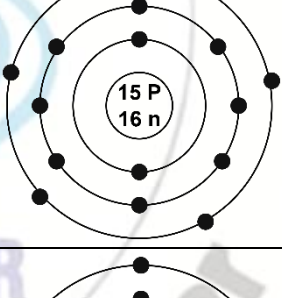
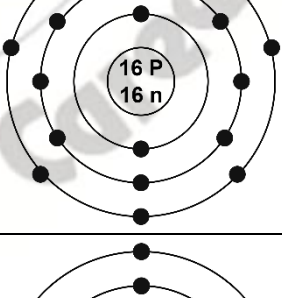
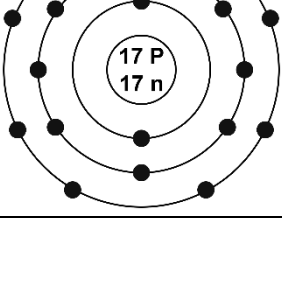
• **Valence Shell**

- The outermost shell of an atom is called valence shell of that atom.
- The electrons present in valence shell is called valence electrons.

▪ **Representation of electrons in shell**

Element	Neutron (n)	e ⁻	Z	Representation
${}^A_1\text{H}^1$ 	$n = A - Z$ $n = 1 - 1$ $\Rightarrow n = 0$	1	1	
${}_2\text{He}^4$	$4 - 2 = 2$	2	2	
${}_3\text{Li}^7$	$7 - 3 = 4$	3	3	
${}_4\text{Be}^9$	$9 - 4 = 5$	4	4	

${}_5\text{B}^{11}$	$11 - 5 = 6$	5	5	
${}_6\text{C}^{12}$	$12 - 6 = 6$	6	6	
${}_7\text{N}^{14}$	$14 - 7 = 7$	7	7	
${}_8\text{O}^{16}$	$16 - 8 = 8$	8	8	
${}_9\text{F}^{19}$	$19 - 9 = 10$	9	9	
${}_{10}\text{Ne}^{20}$	$20 - 10 = 10$	10	10	
${}_{11}\text{Na}^{23}$	$23 - 11 = 12$	11	11	

$_{12}\text{Mg}^{24}$	$24 - 12 = 12$	12	12	
$_{13}\text{Al}^{27}$	$27 - 13 = 14$	13	13	
$_{14}\text{Si}^{28}$	$28 - 14 = 14$	14	14	
$_{15}\text{P}^{31}$	$31 - 15 = 16$	15	15	
$_{16}\text{S}^{32}$	$32 - 16 = 16$	16	16	
$_{17}\text{Cl}^{35}$	$35 - 17 = 18$	17	17	

- **Inert gases / Nobel gases**

- Those atoms which have 8 electrons in their outermost shell are called inert gases.

For example:-

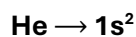
- ✓ Helium (He), Neon (Ne), Argon (Ar), Krypton (Kr), Xenon (Xe), Radon (Rn) etc.

Exception:-

- ✓ Helium (He),

→ Outermost shell $e^-(s) = 2$

Explanation:-



→ Shell no.

- Helium has only 1 shell (K) and only one sub – shell (S) of $2e^-(s)$ i.e, full filled, considered to be a stable.
- Atoms of inert gases are very stable and highly unreactive. So, they can exist in the free state as individual atoms.
- We know that valence electron of an atom takes part in a chemical reaction since inert gases are chemically unreactive, we must conclude that the electron arrangements in their atoms are very stable which do not allow that outermost electrons to take part in chemical reaction.

Note:

- Uses of Inert gases.
 1. Neon is used in advertising sign.
 2. Argon is used in bulb.

Q. Why do atoms combine?

Ans The atoms combine to one another to achieve the inert gas electron arrangement and become more stable i.e, atoms combine to each other for achieve high stability.

- An atom can achieve the inert gas electronic configuration by three ways

1. By losing one or more electrons.
2. By gaining one or more electrons.
3. By sharing one or more electrons.

- **Valency**

- The combining capacity of an atom of an element to form chemical bond is called the valency of that element.
- The valency of an element is decided by the “number of valence electrons” in its atom.
- The valency of an element is either equal to the number of valence electrons in its atom or equal to the number of electrons required to complete eight electrons in the valence shell.
- Any element wants high stability as inert gases. i.e, same outer electronic configuration as inert gases. Thus, any atom either loss or, gain the electrons to achieve electronic configuration as inert gases. So, the number of electrons either gaining or losing is known as valency of that atom.

Note:

- Valency of an atom is equal to either gaining or losing of electrons.

Case 1: Valency of metal

- The valency of a metal is equal to the number of valence electrons in its atom.

$\therefore$ valency of metal = No. of valence $e^-(s)$ in its atom.

- Metals have 1 to 3 electrons in their outermost shell.

exception: { Hydrogen (H) – Valence $e^- = 1$
 Helium (He) – Valence $e^- = 2$

Non – metal

Metal	Valency
Beryllium	2
Sodium	1
Magnesium	2
Aluminum	3
Potassium	1

- Metals having generally “um” suffix.

Note:

1. Metals show maximum valency up to 7 but only valency 1 to 3 is used in the bond formation.

for example:

Element	Valency
Ti	4
V	5
Cr	6
Mn	7
Na	1
Mg	2
Al	3

Like as Outer teeth of Elephant

Like as inner teeth of Elephant

2. Variable Valencies:

- Metal shows variable valencies due to they found in the variable state.

Less valency – US
 High valency – IC } Suffix

Metal State	Valencies
Ferrus	2
Ferric	3
Cuprus	1
Cupric	2
Mercurous	1
Mercuric	2
Argentous	1
Argentic	2
Aurus	1
Auric	2
Stannus	2, 4
Stannic	4
Manganese	2, 4
Platinum	2, 4
Lead → plumbus	2, 4
→ plumbic	2, 4
Phosphorous	3, 5
Platinous	2
Platinic	4

• **Valency of the non – metal**

- The valency of a non – metal is usually equal to eight minus the number of valence electrons in its atom.

∴ Valency of non – metal = 8 – no. of valence electrons in its atom

exception:

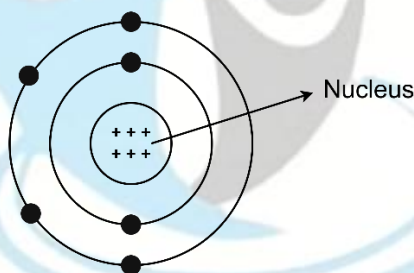
- ✓ valency of hydrogen = 1

for example:-

Non – metals	Valence e ⁻ (s)	valency
Nitrogen	5	8 – 5 = 3
Oxygen	6	8 – 6 = 2
Fluorine	7	8 – 7 = 1
Phosphorus	5	8 – 5 = 3
Sulphur	6	8 – 6 = 2
Chlorine	7	8 – 7 = 1

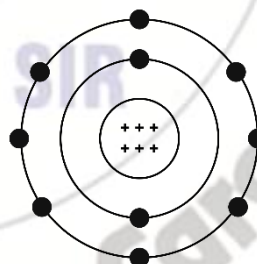
Exception (ii) Carbon → ${}_6\text{C}^{12}$

${}_6\text{C} \rightarrow 1s^2, 2s^2 2p^2$, Valence e⁻(s) = 4



Case 1: When C – atom accepts 4 electrons

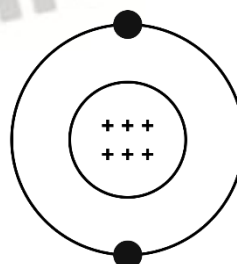
${}_6\text{C} \rightarrow 1s^2, 2s^2 2p^2$



In this case, 6 protons can't hold 10 e⁻(s), Thus it is not possible.

Case 2: When C – atom donates 4 electrons

${}_6\text{C} \rightarrow 1s^2,$



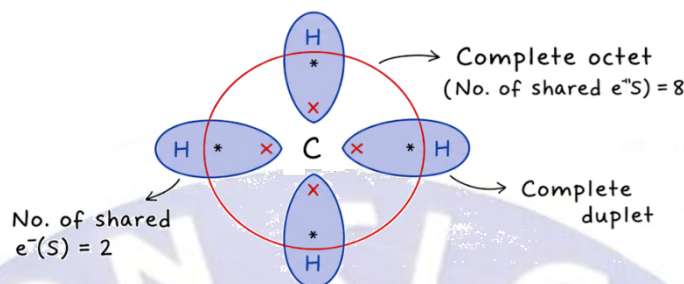
In this case, Six protons of nucleus holding on to just 2 electrons, it can't be possible. In other hand if it loses 4 electrons, it would require a large amount of energy to remove 4 electrons leaving behind a carbon, not possible.

Special Note:

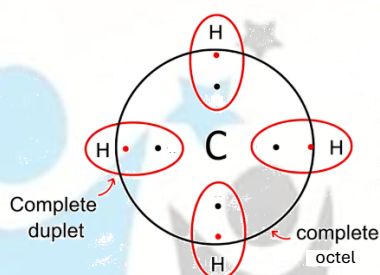
- Carbon can neither gain or loss the electrons it can only share the electrons.

For example:-

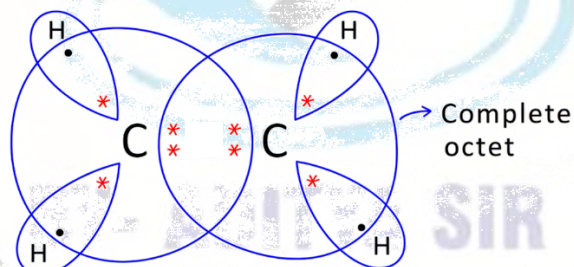
✓ CH_4



✓ CCl_4



✓ C_2H_4

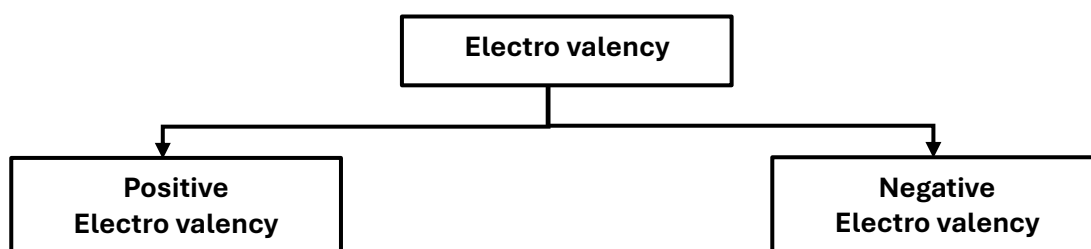


Types of valency

1. Electro valency
2. Covalency

1. Electro valency

- The number of electrons lost or gained by one atom of element to achieve the nearest inert gas electron configuration is known as its electro valency.



a) **Positive Electro valency**

- The elements which lose electrons from positive ions, so they have positive electro valency.

For example:-

- ✓ $_{11}\text{Na} \rightarrow 2, 8, 1$
 $\Rightarrow _{11}\text{Na}^+ \rightarrow 2, 8$ like as $_{10}\text{Ne} \rightarrow 2, 8$
 So, electro valency of Na = +1
- ✓ $_{12}\text{Mg} \rightarrow 2, 8, 2$
 $_{12}\text{Mg}^{2+} \rightarrow 2, 8$
 So, electro valency of Mg = +2 (Like as Neon)
- ✓ $_{13}\text{Al} \rightarrow 1s^2, 2s^2 2p^6, 3s^2 3p^1. [2, 8, 3]$
 $_{13}\text{Al}^+ \rightarrow 2, 8$ Like as Neon configuration
 $\therefore$ electro valency of Al = 3
- ✓ $_{19}\text{K} \rightarrow 1s^2, 2s^2 2p^6, 3s^2 3p^6. [2, 8, 8, 1]$
 $_{19}\text{K}^+ \rightarrow 1s^2, 2s^2 2p^6, 3s^2 3p^6. [2, 8, 8 \text{ as Ar}]$
 Electro valency of K = +1

b) **Negative Electro valency**

- The elements which gain electron from negative ions, so they have negative electro valency.

For example:-

- ✓ $_{7}\text{N} \rightarrow 2, 5 \Rightarrow _{7}\text{N}^{3-} \rightarrow 2, 8$ (Like as Ne)
- ✓ $_{8}\text{O} \rightarrow 2, 6 \Rightarrow _{8}\text{O}^{2-} \rightarrow 2, 8$ (Like as Ne)
- ✓ $_{9}\text{F} \rightarrow 2, 7 \Rightarrow _{9}\text{F}^- \rightarrow 2, 8$ (Like as Ne)
- ✓ $_{15}\text{P} \rightarrow 2, 8, 5 \Rightarrow _{15}\text{P}^{3-} \rightarrow 2, 8, 8$ (Like as Ar)
- ✓ $_{16}\text{S} \rightarrow 2, 8, 5 \Rightarrow _{16}\text{S}^{2-} \rightarrow 2, 8, 8$ (Like as Ar)
- ✓ $_{17}\text{Cl} \rightarrow 2, 8, 7 \Rightarrow _{17}\text{Cl}^- \rightarrow 2, 8, 8$ (Like as Ar)

Here, electro valency of

N & P is -3

O & S is -2

F & Cl is -1

2. **Covalency**

- The number of shared electrons by one atom of an element to achieve the nearest inert gas electron configuration is known as its covalency.

No. of shared e ⁻ (s) of atom	Covalency
1	1
2	2
3	3

For example:-

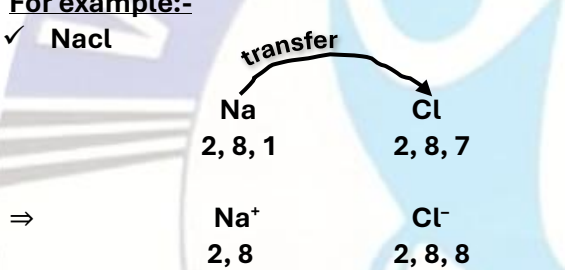
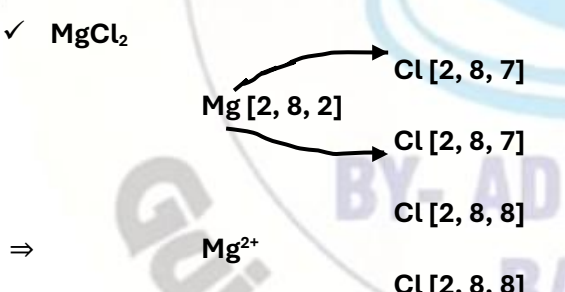
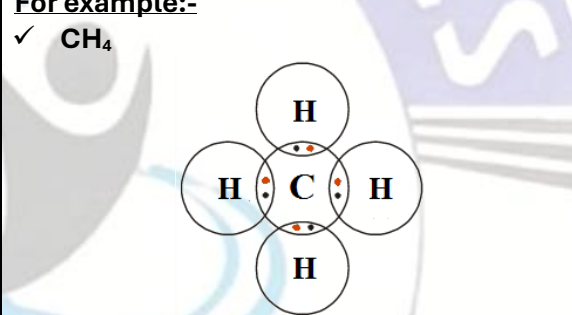
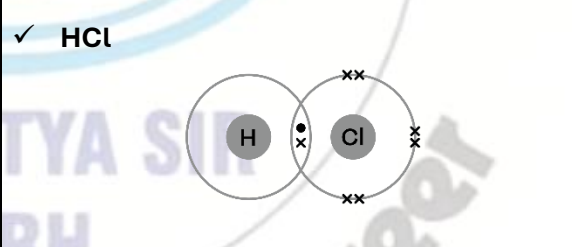
Element	Covalency
Hydrogen	1
Chlorine	1
Oxygen	2

Nitrogen	3
Fluorine	1
Carbon	4
Phosphorus	3
Sulphur	2

Note:

1. The electro valency and covalency of an element are numerically equal.
2. Electro valency is possible for metal as well as non – metal but covalency is only possible in non – metal.

▪ **Differences between electro valency & Covalency**

Electro – valency	Covalency
It is either loss or, gain of valence $e^-(s)$.	It is mutual share of valence electron.
It forms ionic / electrovalent bond.	It forms covalent bonds.
It is either positive or negative.	It is neutral.
This bond is formed between metal & non – metal.	This bond is formed between non – metal and non – metal.
For example:- ✓ NaCl  $\Rightarrow$ ✓ $MgCl_2$  $\Rightarrow$	For example:- ✓ CH_4  ✓ HCl 

Case 3: Valency of Group of Non – metals

- It is defined as, “The sum of valence electrons of one and valency of other”.

For example:-

	Group of Non – metal	Name	Valency
1.	OH	Hydroxide	Valency of O + Valence e^- of Hydrogen $\Rightarrow -2 + 1 = -1$
2.	CO_3	Carbonate	Valence $e^-(s)$ of C + Valence of Oxygen $\Rightarrow 4 + (-2) \times 3$ $\Rightarrow 4 - 6 = -2$

3.	SO ₄	Sulphate	Valence e ⁻ (s) of S + Valency of O ⇒ 6 + (-2) × 4 = -2
4.	PO ₄	Phosphate	Valence e ⁻ (s) of P + Valency of O ⇒ 5 + (-2) × 4 = -3
5.	NO ₃	Nitrate	Valence e ⁻ (s) of N + Valency of Oxygen ⇒ 5 + (-2) × 3 = -1
6.	Cr ₂ O ₇	Dichromate	2 × 6 + (-2) × 7 = -2
7.	CrO ₄	Chromate	6 + (-2) × 4 = -2
8.	HCO ₃	Hydrogen Carbonate	1 + (-2) = -1
9.	SO ₃	Sulphite	-2
10.	NH ₄	Ammonium	-3 + 4 = 1
11.	MnO ₄	Manganate	7 + (-2) × 4 = -1
12.	MnO ₄	Permanganate	-2
13.	HCO ₃	Bicarbonate	1 + (-2) = -1
14.	NH ₃	Ammonia	5 - 3 + 3 × 1 = -3 + 3 = 0

• **Chemical / Molecular Formula**

- The chemical formula of a compound represents the composition of a molecule of the compound in terms of the symbol of the elements present in it.

• **Writing of formulae of molecular compound**

- If we know the valencies of elements, then we can work out the formulae of their compounds by balancing the valencies of the different atoms which occur in the compound.

Note:

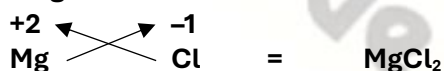
- In writing the chemical formula, we do not write the charges on the symbol of ions.

▪ **Formation of molecules**

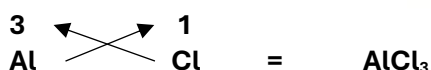
1. Sodium Chloride → NaCl



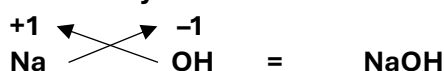
2. Magnesium Chloride → MgCl₂



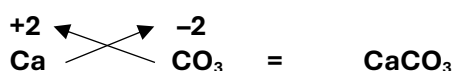
3. Aluminum Chloride → AlCl₃

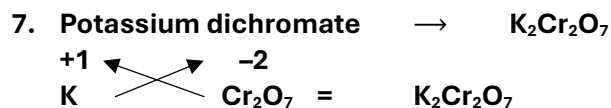
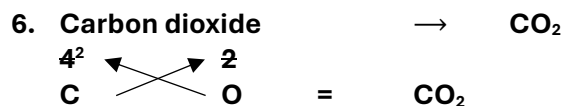


4. Sodium Hydroxide → NaOH



5. Calcium Carbonate → CaCO₃

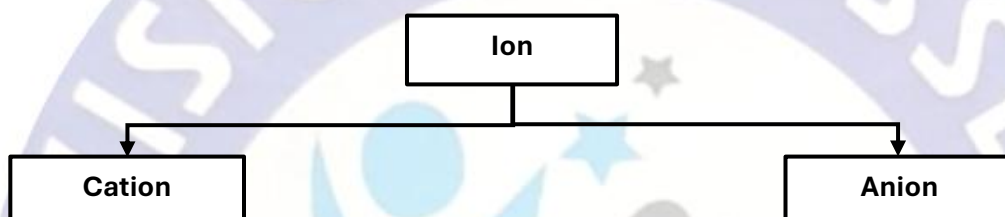




IONS

• Definition

- An electrically charged atom or molecule formed by loss or gain of electrons is called ions.
- It is the charged particle / atom.



1. Cation

- The ion of positive electrical charge is called cation.
- Metal forms cation to donate the electrons.
- Cation has no. of equal positive charge as no. of donating electrons.

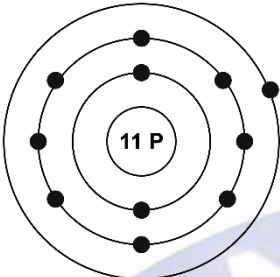
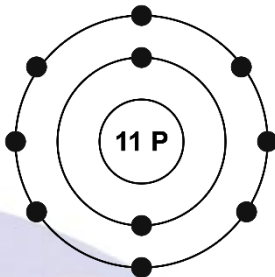
For example:-

Metal	Its cation
Li	Li^+
Be	Be^{+2}
Na	Na^+
Mg	Mg^{2+}
Al	Al^{3+}
K	K^+
Ca	Ca^{2+}
Sc	Sc^{+1}

Note:

- Monovalent Cations $\rightarrow$ Valency + 1
- Divalent Cations $\rightarrow$ Valency + 2
- Trivalent Cations $\rightarrow$ Valency + 3

▪ **Differences between metal and its cation**

Metal (Na)	Its Cation (Na ⁺)
Electronic configuration of Na = 2, 8, 1 	Electronic configuration of Na ⁺ = 2, 8 
Its size is larger due to three shells.	Its size is smaller due to two shell.
It is electrically neutral.	It is electrically positive.
It is less stable as compare to its cation.	It is more stable as compared to its atom.
It has more no. of electrons in its shell.	It has less no. of electrons in its shell.
It has three shell (K, L, M)	It has two shell (K, L)

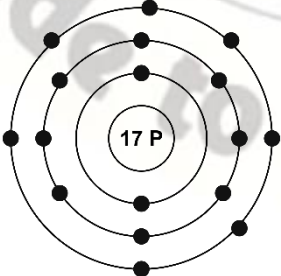
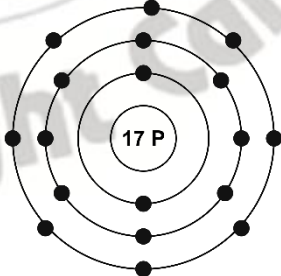
2. **Anion**

- The ion of negative electrical charge is called anion.
- Non – metals form anion to gain e⁻(s).
- Anion has equal no. of negative charge as the no. of gaining e⁻(s).

For example:-

Non – metal	Its anion
Nitrogen	N ³⁻
Oxygen	O ²⁻
Fluorine	F ⁻¹
Phosphorus	P ⁻³
Sulphur	S ⁻²
Chlorine	Cl ⁻¹

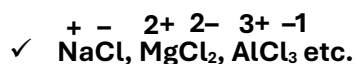
▪ **Differences between non – metal and its anion**

Non – metal (Cl)	Anion (Cl ⁻)
$_{17}\text{Cl} \rightarrow 2, 8, 7$ 	$_{17}\text{Cl}^- \rightarrow 2, 8, 8$ 
It is smaller in size.	It is larger in the size.
It has less no. of e ⁻ (s) as compared to its anion.	It has more no. of e ⁻ (s) as compared to its original atom.
It is less stable.	It is high stable.
It is electrically neutral.	It is electrically negative.

Note:

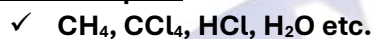
1. Cation and anion both together form ionic compound.

For example:-



2. Non – metal and non – metal together form covalent compound.

For example:-



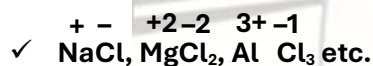
- 3.

	Anion	Valency
1.	Monovalent ($\text{F}^{-1}, \text{Cl}^{-}, \text{Br}^{-}, \text{I}^{-}$)	-1
2.	Divalent ($\text{O}^{-2}, \text{S}^{-2}$)	-2
3.	Trivalent ($\text{N}^{-3}, \text{P}^{-3}$)	-3

• **Radical**

- The charged particles in ionic compounds are called radicals.

For example:-



For example:-

